

PARKTOWN BOYS' HIGH SCHOOL



Department of Mathematics

Cycle Test

25 March 2026

Subject	Mathematics
Grade	12
Topic	Calculus
Marks	50
Time	1 hour

Examiner

Mr E Maine

Moderator

Mrs S Rowell

INSTRUCTIONS

Read the following instructions carefully before answering the questions.

1. This question paper consists of 4 questions and 4 pages. Please ensure that your question paper is complete.
2. Answer ALL the questions.
3. Clearly show ALL calculations, diagrams, graphs, etc. that you have used in determining your answers.
4. Diagrams are NOT necessarily drawn to scale.
5. Number your questions exactly like the questions in the question paper.
6. Answers only will NOT necessarily be awarded full marks.
7. You may use an approved scientific calculator (non-programmable and non-graphical), unless stated otherwise.
8. If necessary, round off answers to TWO decimal places, unless stated otherwise.
9. Use the mark allocation to guide the length of your solution.
10. Write neatly and legibly.
11. Rule off after each question.

QUESTION 1

1.1 If $f(x) = -\frac{2}{x}$, determine $f'(x)$ from first principles. (5)

1.2 Determine:

1.2.1 $\frac{dy}{dx}$, if $y = 16x^2 + \frac{3}{x^{10}}$ (3)

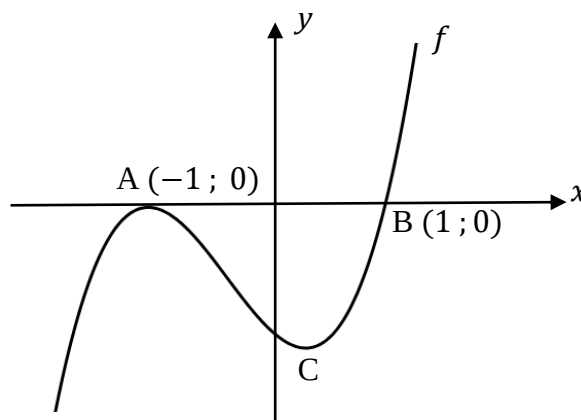
1.2.2 $D_x \left[x^2 \left(3x - \sqrt[3]{x^2} \right) \right]$ (3)

1.2.3 $f'(x)$ if $f(x) = \frac{8 - 27x^3}{2 - 3x}$ (4)

[15]

QUESTION 2

The sketch below shows the graph of $f(x) = x^3 + px^2 + qx + r$. A(-1;0) and B(1;0) are the x – intercepts of f . A and C are stationary points of f .



2.1 Show that $p = 1$, $q = -1$ and $r = -1$. (3)

2.2 Determine the coordinates of C. (4)

2.3 Determine the equation of the tangent to f at $x = 2$. (3)

2.4 Determine the x – coordinate of the point where the tangent from 2.3 touches the graph of f again. (4)

2.5 For which values of x will $f(x)$ be concave down? (2)

[16]

QUESTION 3

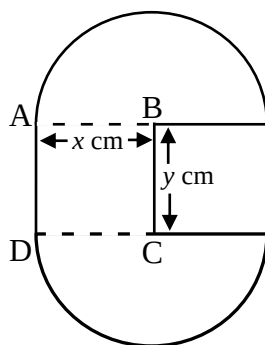
Draw a sketch of $f'(x)$, showing all relevant information, if:

- $f(x)$ is a cubic function with turning points at $x = -2$ and $x = 4$.
- $f''(x)$ is positive at $x = -5$.

[3]

QUESTION 4

- 4.1 The diagram shows a badge made of thin sheet metal, consisting of two semi-circular pieces with centres B and C and radius x . The semi-circles are connected by a rectangular section, ABCD, such that $AB = x$ and $BC = y$. The area of the badge is 20 cm^2 .



- 4.1.1 Show that the perimeter, P , of the badge is given by: $P = 2x + \frac{40}{x}$. (3)
- 4.1.2 Calculate the minimum perimeter of the badge. (4)
- 4.2 An experiment is conducted to observe the motion of a particle. The particle starts from rest at point A and moves until it comes to rest again at point B. The velocity of the particle (in m/s) is represented by $s'(t) = -3t^2 + 12t$.
- NOTE: velocity is the rate of change in distance with respect to time.
- 4.2.1 Show by calculation that the particle is speeding up after exactly 1 second. (2)
- 4.2.2 Calculate the maximum speed the particle reached. (3)
- 4.2.3 Calculate the distance between points A and B. (4)

[16]